Xi'an Jiaotong-Liverpool University
西交利物浦大学

Longitudinal Metagenetic Analysis of Microbial Transmission Dynamics and Antibiotic Resistance Gene Dissemination between Hand and Residential Surfaces

BIO303 FINAL YEAR PROJECT

FINAL REPORT

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摘要

背景：手作为微生物的传播媒介，能够促进病原菌在室内表面的传播和扩散。本研究采用纵向宏基因组学方法调查了手部微生物群与居住环境表面的动态传播模式，并关注其所携带的抗生素耐药基因（ARGs）的传播。

方法：我们基于健康个体手部和住宅表面的高通量测序数据对基因组进行了重建和注释。Bray-Curtis 相异度和中兴群落模型被用于研究微生物群落的 beta 多样性和扩散能力。此外，我们利用 MobileOG、Resfams 和 CARD 数据库鉴定了基因组中的可移动遗传元件(MGEs)、抗生素耐药蛋白和基因。

结果：在群落水平上，我们发现手和门把手表面的微生物群落出现了同质化现象。在物种水平上，我们观察到 *Gordonia bronchialis*、*Janibacter indicus* 和 *Micrococcus luteus* 具有最高的丰度。在基因组水平上，我们观察到 8 个含有抗生素耐药蛋白和插入序列的基因组在手和门把手表面上具有明显的传播迹象。此外，基因组中 ARGs 和 MGEs 的共现暗示了 ARGs 在微生物物种之间存在潜在的水平基因转移(HGT)。

讨论：本研究为手部和住宅表面之间的微生物交换提供了见解，使我们能够在基因组水平上研究室内微生物的传播及其相关的 ARGs。对病原微生物的传播模式和 ARGs 的 HGT 的进一步研究有助于未来开发模型和干预措施，以减少负面影响，增强健康和福祉。

关键词：鸟枪法宏基因组学，手部微生物组，传播潜力，抗生素耐药基因，可移动元件

Abstract

Introduction: As a microorganism-transmission medium, hands can facilitate the transmission and dispersal of pathogens on indoor surfaces. This study adopted longitudinal metagenomics to investigate the dynamic transfer patterns of microorganisms between the hand and residential surfaces, and their antibiotic resistance genes (ARGs) at the genomic level.

Methods: Genome reconstruction and annotation was performed using high-throughput sequencing data from a healthy individual's hand and the residential surfaces. Bray-Curtis dissimilarity and Sloan Neutral Community Model were used to study the community discrepancy and dispersal capability of microorganisms between hand and the doorknob. MobileOG, Resfams, and CARD databases were used to identify mobile genetic elements (MGEs), antibiotic-resistant proteins and genes in the genomes.

Results: At the community level, we observed the homogenization of the microbial communities on the hand and door knob surfaces. At the species level, we observed that *Gordonia bronchialis*, *Janibacter indicus*, and *Micrococcus luteus* were the most prevalent species. At the genomic level, eight species that displayed distinct transmission patterns between hand and doorknob over time, containing antibiotic resistance proteins and insertion sequences. The co-occurrence of ARGs and MGEs indicate the potential horizontal transfer history of ARGs between microbial species.

Discussion: This study provides insights into the microbial exchange between hands and residential surfaces, allowing us to investigate indoor microbial transfer and its associated ARGs at the genome level. Further research on transmission patterns of pathogenic microorganisms and horizontal gene transfer of ARGs contributes to the future development of models and interventions to reduce negative outcomes and enhance the health and well-being of human occupants in built environments.

Keywords: shotgun metagenomics, hand microbiome, transmission potential, antibiotic resistance gene, mobile genetic element

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Introduction

The commensal skin microbiota consists of communities of bacteria, fungi, viruses, and parasites. As the largest organ in the human body, skin plays a crucial role in the maturation and homeostatic control of the host immune network (Belkaid and Tamoutounour, 2016, Elias, 2007, Proksch *et al.*, 2008, Prescott *et al.*, 2017). Due to global urbanization, the majority of human life is spent in constructed habitats (Klassert *et al.*, 2021). The transmission of microbial communities through skin-to-skin contact is especially prominent in indoor environments, where human skin frequently encounters surfaces that may harbor microorganisms (Klassert *et al.*, 2021, Horve *et al.*, 2020). Leung *et al.* (2018) observed the presence of distinct microbial communities in interior environments associated with individual and household characteristics. Lax *et al.* (2014) discovered that the indoor flora of the residence was similar to that of its occupants' skin, particularly their hands.

However, the transition of commensal bacteria into pathogens may be facilitated by dysbiosis, genetic variation in the host, and altered immune status. Antibiotic resistance genes (ARGs) may also develop when pathogens are exposed to various selective pressures, including antibiotic therapy, nutrition, and lifestyle (Baron *et al.*, 2018, de Abreu *et al.*, 2020). Microbial community exchange in the interior environment may contribute to the propagation of ARGs (Hyatt *et al.*, 2010). Mobile genetic elements (MGEs) in bacteria may contribute to the transmission of ARGs from non-pathogenic bacteria to pathogenic bacteria through frequent horizontal gene transfer (HGT) between conspecific genera, which increases the likelihood of multi-drug resistant bacteria infection (Groussin *et al.*, 2021, Ellabaan *et al.*, 2021, Jiang *et al.*, 2017, von Wintersdorff *et al.*, 2016, Wybouw *et*

al., 2016). Consequently, the development, acquisition, and transmission of ARGs in the indoor environment are crucial factors influencing antimicrobial resistance on human skin and the entire microbial community. Most of the previous research on ARGs in the human microbiome, however, was limited to the intestine (Devirgiliis *et al.*, 2011, Modi *et al.*, 2014, Hu *et al.*, 2013, Rolain, 2013). The antibiotic resistance profile of the cutaneous microbiome is poorly understood.

Recent advances in high-throughput DNA sequencing of human-associated microbiome collections have allowed scientists to define the distribution of dominant bacteria and ARGs on the hand surface (Bahram *et al.*, 2018, Baron *et al.*, 2018, Danko *et al.*, 2021, Hendriksen *et al.*, 2019, Hu *et al.*, 2013, Kang *et al.*, 2018). In 2018, Wilkins *et al.* (2021) initiated a 10-day microbial survey of four unconnected residential units and their surrounding outdoor environment using shotgun metagenomic sequencing. They demonstrated a high degree of accuracy in matching microbial communities to individuals and described diurnal changes in the skin and surface bacteria (Wilkins *et al.*, 2021). However, their study characterized the microbial transmission at the community level (low resolution) rather than genome level (high resolution).

Therefore, this study aims to examine one of the healthy individual and the corresponding residential environment, focusing on the longitudinal metagenomic analysis to reveal the dynamic microbial transmission patterns on the skin and residential surface at the genome level, and to investigate the carriage and diffusion of ARGs during microbial transmission.

Methods and Materials

In this study, a series of metagenomic data processing and analysis were carried out on the high-throughput sequencing data, and the analysis process is shown in **Figure 1**.

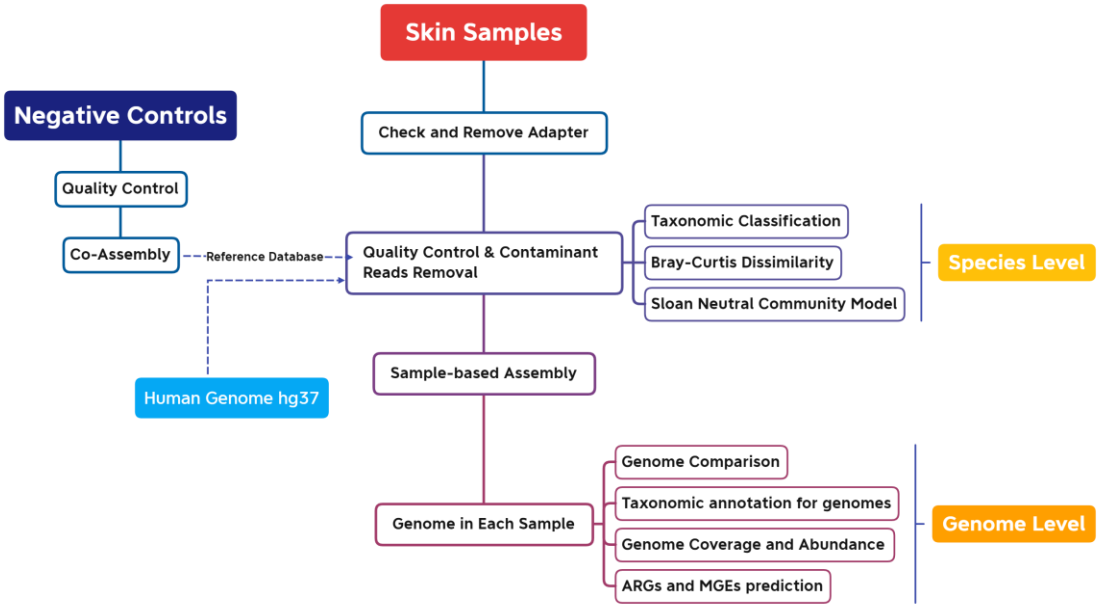

Figure 1. Workflow of Data Analysis on negative controls and Skin Samples

Data Collection

Swab samples from four individual residential units were collected from the skin (left and right palms) and indoor surfaces (headboards and doorknob) twice daily (8-9 am, 5-8 pm) for 10 days in May 2018. The four participants in the study lived independently and followed their customary daily routines throughout the research. Following DNA extraction from swab samples, Illumina 150 bp paired-end sequencing was performed. Due to the high quality and completeness of residence 3 data, the consideration of available computing resources, and the accurate descriptions of individual change patterns in longitudinal studies, only residence 3 was analyzed. The data contains the paired-end

sequence of 12 negative controls and 59 skin (left and right palm) and bedroom interior doorknobs from residence 3, downloaded from NCBI Bioproject (Accession: PRJNA671748). Headboard samples were excluded from the analysis because the headboard surface is less frequently touched by hand with few microbial transfers from the immediate hand surface.

Quality control and Decontamination

Fastqc (v0.11.9) was first utilized to examine the sequence's adaptor (Andrews, 2010), while MultiQC (v1.7) was utilized to view the combined report. AdapterRemoval (v2.3.3) was used to remove adaptors from sequences for specified sequences (Schubert *et al.*, 2016). To eliminate contaminated reads, MetaWRAP (v1.3.2) was used to co-assemble the 12 quality-filtered negative control reads using the megahit (v1.1.4) approach with the minimum contig length of 1000 bp (Uritskiy *et al.*, 2018). KneadData (v0.12.0) with Bowtie2 (v2.4.2) (Langmead and Salzberg, 2012) and Trimmomatic (v0.39) (Bolger *et al.*, 2014) was used to perform filtering and dressing quality and remove the human sequence and contaminant sequences with human genome hg37 and co-assembled control contig file as the references.

Bray-Curtis Dissimilarity Analysis and Principal Coordinates Analysis

All downstream analyses were performed using R (v4.2.3) (R Core Team, 2023). vegdist function in the R package vegan(v2.6.4) (Oksanen *et al.*, 2013) was used to calculate dissimilarity indices based on the "bray" method, respectively. Bray-Curtis line plots were drawn using ggplot2 package (v3.4.2) (Villanueva and Chen, 2019). Then, cmdscale() argument in R was used to perform principle coordinates analysis (PCoA) (Gower, 1966) and variance were calculate by eigenvals function in vegan package

(v2.6.4)(Oksanen *et al.*, 2013). In addition, the AICcPermanova package provided an R implementation of model selection for Permanovas from the vagan package and calculated AICc. The PCoA images were plotted with stat_ellipse function for a multivariate t-distribution by ggplot2 package (v3.4.2) (Villanueva and Chen, 2019).

Taxonomy classification

Paired-end reads were taxonomically annotated against the MiniKraken_v1_8GB database (April 2019 version) using Kraken2 (v2.0.7-beta)(Wood *et al.*, 2019). The Kraken2 output report files were consolidated using the Python (v3.7) script kraken-multiple.py and the taxa information was determined using the Python script kraken-multiple-taxa.py.

Sloan Neutral Community Model

To fit the observed abundance and frequency to a beta distribution derived from neutral theory, R (v4.2.3) (R Core Team, 2023) was used to build a neutral community model (NCM). The average relative abundance and occurrence frequency of each taxa across communities were calculated according to methods described by (Sloan *et al.*, 2006) and (Burns *et al.*, 2016). The parameters of Binomial model (AIC_bino =-5118.79） and Poisson model (AIC.pois = -5118.79) were fitted using Non-linear least squares (NLS) and Maximum likelihood estimation (MLE). Sloan neutral model images were used using the ggplot2 package (v3.4.2) (Villanueva and Chen, 2019). Global Catalogue of Pathogens (gcPathogen) database (https://nmdc.cn/gcpathogen/), the bacterial taxa are downloaded directory is used to retrieve samples of pathogens.

Genome Assembly and Comparisons

Sample-based clean true samples were assembled using megahit (v1.1.4) from MetaWRAP (v1.3.2). Because the sequences of the two pairs do not correspond to each other when the sequences of the Fastq files are processed separately with Python, the repair.sh argument in BBmap (v39.01) is used to repair the fastq file (Wood *et al.*, 2019). The repaired fastq was then passed through MetaWRAP (v1.3.2) built-in metabat2 (v2.15) (Kang *et al.*, 2019), maxbin2 (v2.2.7) (Wu *et al.*, 2016), concoct(1.1.0)(Alneberg *et al.*, 2013) for binning. In addition, CheckM(v1.0.13) (Parks *et al.*, 2015) with hmmer(v3.3.2), prodigal(v2.6.3) (Hyatt *et al.*, 2010), The lineage-specific approach of pplacer(v1.1) was used to select genomes with a completeness ≥ 50 and a contamination level ≤ 20. Finally, the optimal 186 metagenomic assembled genomes (MAGs) were obtained from the 53 sample assemblies.

To remove redundant genomes, dRep (v3.0.0) was used to dereplicate MAGs, with an average nucleotide identity of > 99% considered as identical genomes, resulting in the representative MAGs (rMAGs). The taxonomic annotation of rMAGs was performed using the classify_wf option of GTDB-TK (v2.1.1) (Chaumeil *et al.*, 2020) annotate the MAGs in the sample for species classification based on the GTDB reference tree (release207_v2). Subsequently, CoverM (v0.6.1) was used to calculate the coverage of rMAGs in short reads. Subsequent plots were performed using the ggplot2 package (v3.4.2) (Villanueva and Chen, 2019).

Gene prediction and Annotation

Prokka (v1.14.6) was used to convert nucleotide sequence (.fa format) into protein sequences (.faa format) before functional annotation. Resfams (v1.2) was used to confirm

antibiotic resistance function in rMAGs (Gibson et al., 2015). The ggplot2 package (v3.4.2) (Villanueva and Chen, 2019) was used to draw the stack map of resistance gene classification. Subsequently, to analyze the presence of MGEs on rMAGs, ISEScan (v1.7.2.1) was used to identify insertion sequences in the prokaryotic genome. The entire contigs and inserts found were further compared with the mobileOG (beatrix-1.6) database (value threshold 1e-20) and Comprehensive Antibiotic in proksee (v1.0.0a6) Resistance Database (CARD; https://card.mcmaster.ca) (Alcock et al., 2020) were aligned and plotted to visualize MGEs and ARGs.

Results

Bray-Curtis dissimilarity between palms and doorknob

The results of the comparison of Bray-Curtis dissimilarity (BCD) at different time intervals (time between collection of reference samples and query samples) are shown in **Figure 2A**. When the time interval is negative (the query sample is collected before the reference sample), the shorter the time interval from 0, the smaller the BCD. The BCD is minimal when the time interval approaches 0. When the time interval is positive (the query sample is collected after the reference sample), the BCD first grows and then decreases with the increase of the time interval. In addition, the PCoA results for group composition based on sample type (**Figure 2B**) revealed that there was no significant difference in microbial taxa between palm and doorknob (ANOSIM $p = 0.014$, $R^2 = 0.07$). As microbes proliferate, these findings may suggest a gradual homogenization of microbial taxa on hands and doorknob. However, the areas of the PCoA map that did not overlap, i.e., those

taxa with low similarity between doorknob and palm, may be because certain microorganisms did not disseminate or declined during transmission.

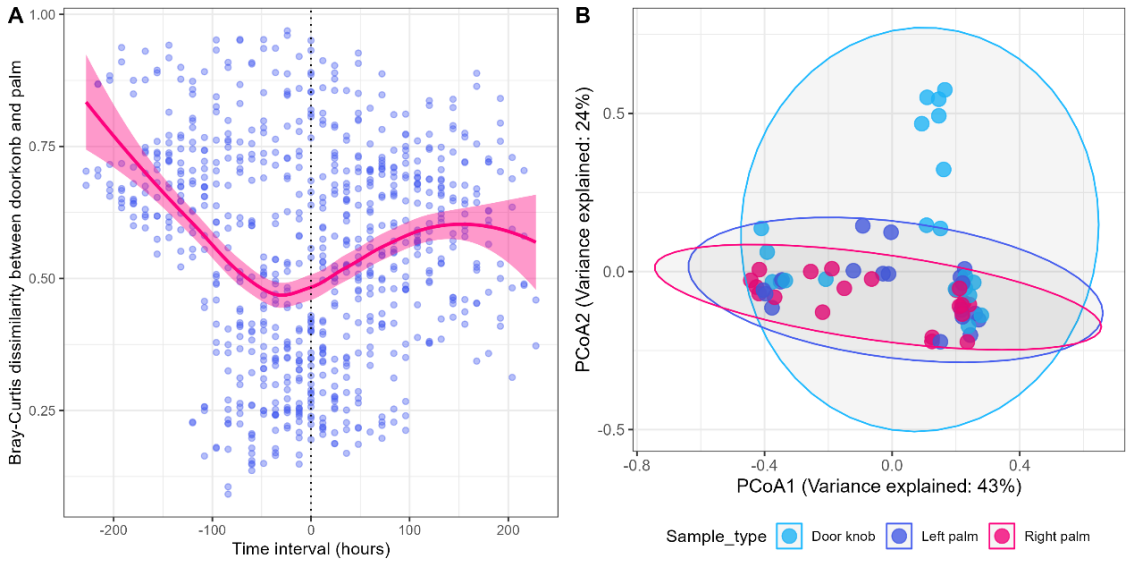

Figure 3. Bray-Curtis dissimilarity and Principal coordinate analysis (PCoA) between doorknob and palms. (A) *Bray-Curtis line plot. Each point in the scatter represents the proportion of query sample matching attempts that correspond to the correct location at each time point. Locally estimated scatter plot smoothing moving averages and 95% confidence intervals are represented by pink lines and shaded areas, respectively. The time interval is the approximate duration (in hours) between the collection of the reference samples and the query samples.* ***(B)*** *Bray-Curtis PCoA by sample type. Blue, purple, and pink dots represent groups of organisms on the doorknob, left palm, and right palm, respectively. Circles and shadows within the circles represent 95% confidence intervals. PCoA1 (x-axis) and PCoA2 (y-axis) respectively explained 43% and 24% of the variance of the microbial taxa amongst the different sample type.*

Taxonomy Classification

The microbial communities in each sample were investigated, and all identified microorganisms belonged to domain bacteria. The distribution of the top 12 most abundant

taxa across the samples at different times is shown in **Figure 3**. *Gordonia bronchialis* (22.69±13.03%), *Janibacter indicus* (19.31±18.90%) and *Micrococcus luteus* (5.31±7.35%) were the three most abundant bacterial species. Most of these bacteria belonged to the the Micrococcales and Mycobacteriales orders of Actinomycetia class, both of which are prevalent species on the skin.

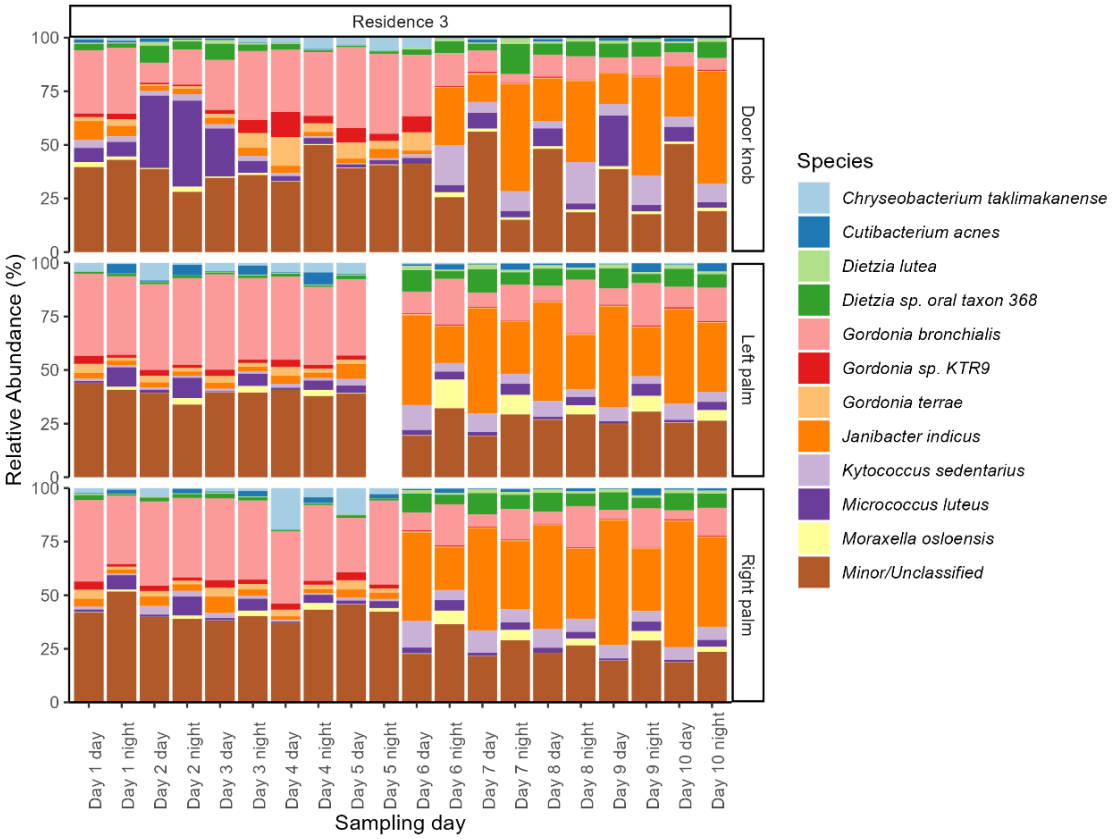

Figure 3. Species composition and relative abundance of doorknob and palm at the species level. *The X-axis represents sampling time and the Y-axis represents relative abundance (%). The top 11 most abundant species are shown in different colors, and the remaining readings are classified as "other/unclassified". Vacancy values are sequence deletions during sequencing. The g represents the genus level and s represents the species level.*

Dispersal event of microbial taxa from hand to the doorknob

NCM was used to demonstrate the dispersal events of microbial taxa occurring between the hand and the indoor surfaces (**Figure 4**). As predicted by the Sloan model, 77% of taxa during dispersal from hand to doorknob were in the neutral interval of the model, and they were considered to have normal dispersal capacity. Taxa above the model-predicted frequency (21%) meant that they were more likely to disperse, even if they had lower abundance. Conversely, taxa below the model-predicted frequency (2%) had deficient dispersal capacity. In addition, pathogens were identified in different partitions of the Sloan model, accounting for 5.75%, 6.10%, and 16.48% of the classification of the partition, respectively (**Table 1**). Among them, 47 pathogens in the above partition had strong transmission ability, including some common skin bacterial pathogens (**Appendix Table 1**).

Table 1. The proportion of pathogens identified on different partitions

Partition	Number of Species	Number of Pathogen
Above (5.75%)	818	47
Neutral (6.10%)	2966	181
Below (16.48%)	91	15

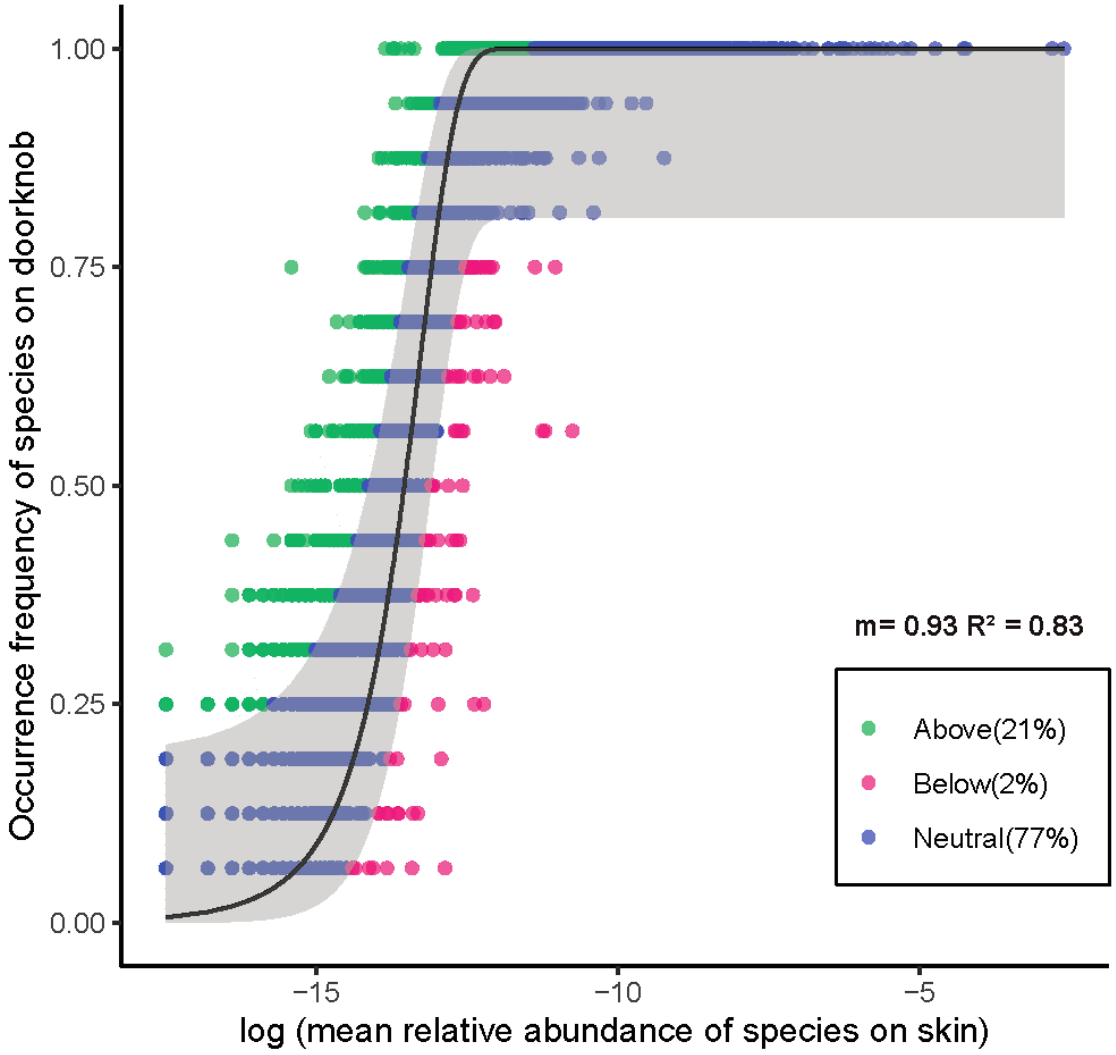

Figure 4. Sloan neutral community model (NCM) for the dispersal of microbial taxa from hands to doorknob. *The x-axis represents the log value of the mean relative abundance of the species on the skin, while the y-axis represents the frequency of occurrence of the species on doorknob. The black line represents the best fit and the shaded areas indicate the 95% confidence intervals predicted. Blue, green, and red dots represent species with a dispersal ability within, above, or below model prediction, respectively. The m represents the migration rate. R^2 indicates the fitness of the model.*

Microbial transmission pattern at the genome level

All species were identified as bacteria at the genome level. Among them, nine microbial genomes were found on both palms and doorknobs (**Figure 5**). The results revealed that the frequency of microbial genomes recovered on doorknobs was lower than on palms and most microbial genomes were first detected on palms. Meanwhile, **Figure 6** illustrates the coverage of rMAGs of these nine genomes in each sample. It was found that these genomes might be present in the reads with varying degrees of coverage within the ten days but were not identified due to poor quality of their genomes.

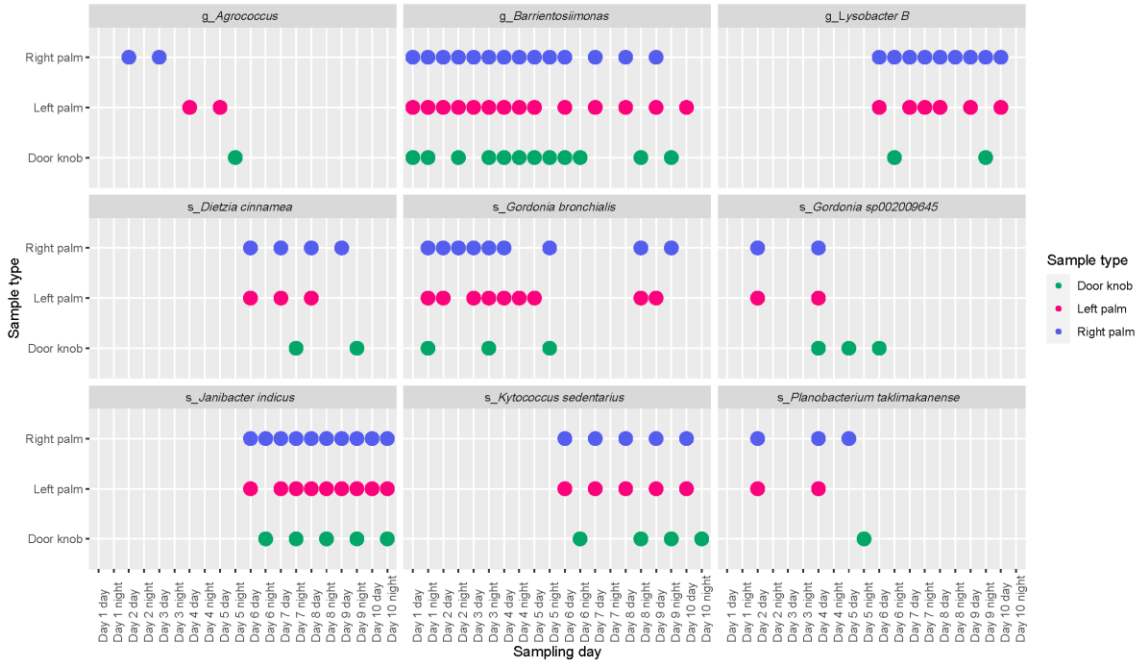

Figure 5. Microbial transmission pattern at the genome level. *Blue, red, and green dots represent the presence of this microbial genome in the right palm, left palm, and doorknob, respectively. In the title of each panel, g represents the genus level, and s represents the species level.*

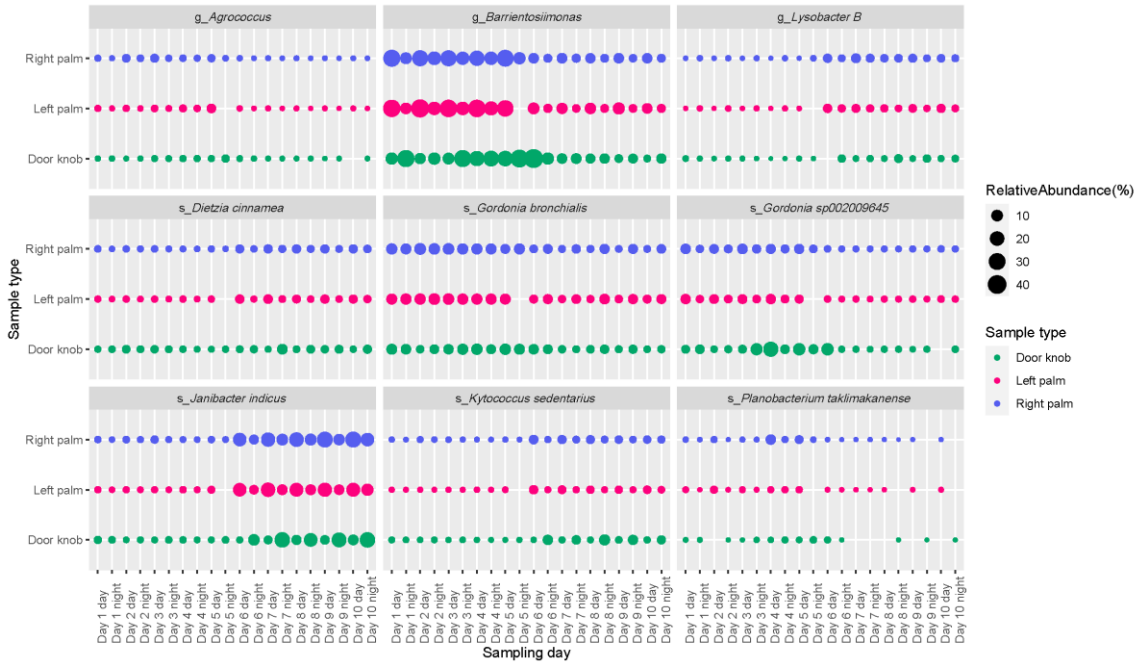

Figure 6. Coverage of rMAGs in each sample. *Blue, red, and green dots represent the occurrence of rMAGs on the reads of the right palm, the left palm, and the doorknob, respectively. The size of the dots represents the relative abundance (%). The g represents the genus level and s represents the species level.*

Identification of antibiotic protein families

Twelve antibiotic protein families were identified from the above rMAGs considered as potentially transmissible (**Figure 7**). Among them, the most prevalent antibiotic resistance mechanism was ATP-binding cassette (ABC) antibiotic efflux pump, which was found in all eight rMAGs across skin and doorknob samples. Other protein families that occur across multiple genomes include MFS major facilitator superfamily (MFS) antibiotic efflux pump, tetracycline resistance MFS efflux pump, and resistance-nodulation-cell division (RND) antibiotic efflux pump are additional protein families found in multiple genomes. The remaining protein families were exclusive to each individual rMAG.

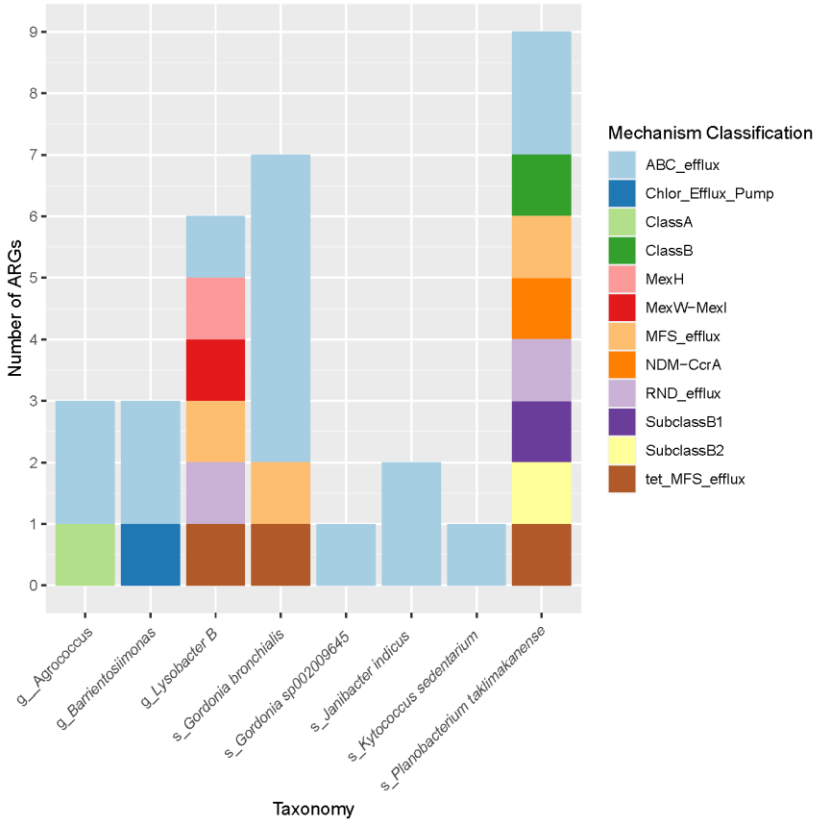

Figure 7. Antibiotic resistance of rMAGs. *The families of antibiotic resistance protein were color-codeds. Abbreviation: ABC_efflux: ATP-binding cassette (ABC) antibiotic efflux pump; Chlor_Effluex_Pump: chloramphenicol efflux pump; Class A: Class A beta-lactamase; Class B: Class B beta-lactamase; MexH: membrane fusion protein of the efflux complex MexGHI-OpmD; MexW-MexI: A grouping of related mexW and mexI subunits of efflux pumps conferring antibiotic resistance; MFS_efflux: major facilitator superfamily (MFS) antibiotic efflux pump; NDM_CcrA: A grouping of related NDM and CcrA beta-lactamases; RND_efflux: resistance-nodulation-cell division (RND) antibiotic efflux pump; Subclass B1: Subclass B1 (metallo-) beta-lactamase hydrolize penicillins, cephalosporins and carbapenems; Subclass B2: Subclass B2 (metallo-) beta-lactamase selectively hydrolize carbapenems; tet_MFS_efflux: tetracycline resistance MFS efflux pump.*

Mobile Genetic Elements and Antibiotic Resistance Genes Identification

The identification of MGEs and ARGs on the genome showed that MGEs and ARGs co-occurred only on rMAGs of genus *Lysobacter B* (**Figure 8**). However, there was no overlap between their regions.

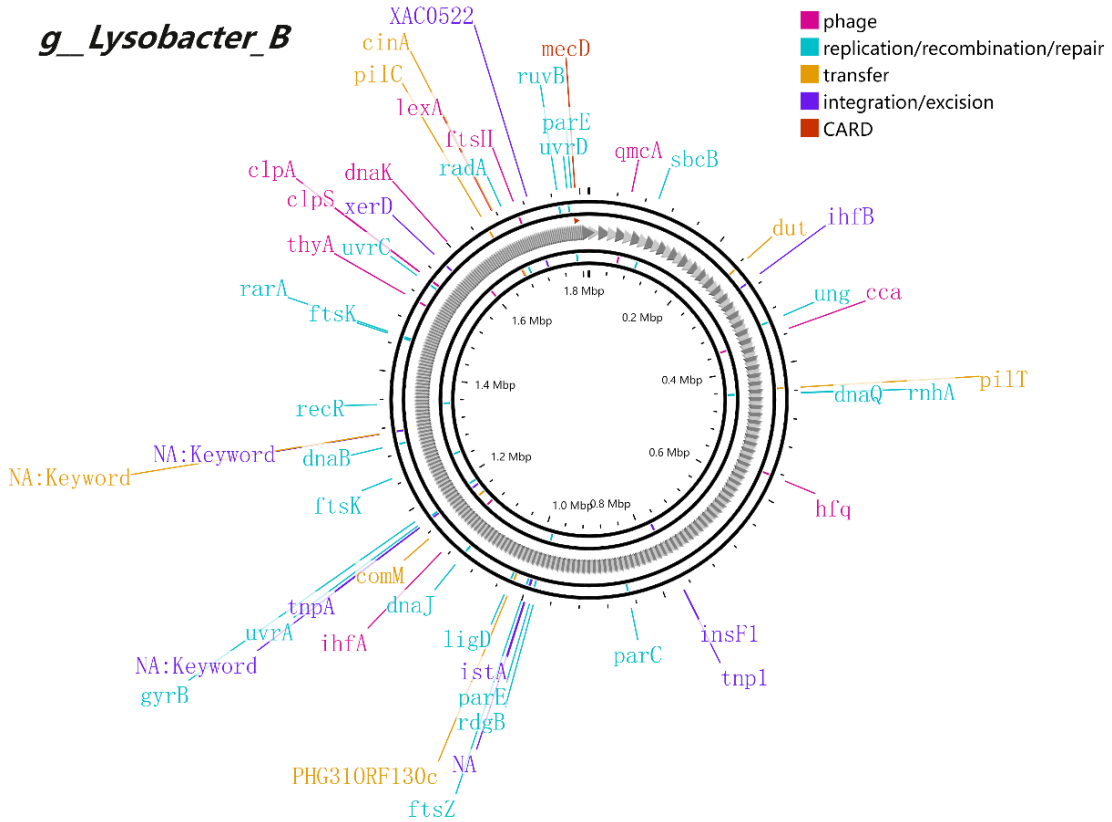

Figure 8. MGEs and ARGs of Lysobacter B rMAGs. *MGEs include phage, replication/recombination/repair, transfer, integration/excision, Label with different colors. ARGs identified from the CARD database are marked in red.*

Insertion sequences (IS) are the smallest but most abundant mobile elements in the prokaryotic genome. Thus, 104 IS were identified from the eight rMAGs (**Table 2**), occupying 0.19% to 1.3% of each genome. The results of further retrieval of resistance genes in IS (**Figure 9**) showed that ARGs might exist in every segment of IS.

Table 2. Insertion Sequence detected in contigs

Taxonomy	nIS	%Genome	bps4IS	dnaLen
s_*Planobacterium taklimakanense*	20	1.3	33659	2596718
s_*Gordonia bronchialis*	30	0.8	36098	4530003
g_*Agrococcus*	8	0.77	14146	1841737
s_*Janibacter indicus*	14	0.59	17087	2897137
g_*Barrientosiimonas*	11	0.44	15718	3543856
s_*Kytococcus sedentarius*	8	0.35	6887	1995664
g_*Lysobacter B*	6	0.26	4624	1812744
s_*Gordonia sp002009645*	7	0.19	7467	3872203

nIS:the number of IS copies; %Genome: the percentage of the genome spanned by IS sequences; bps4IS: length of sequence segment spanned by the IS sheet in the sequence jdnaLen: total nucleic acid length of the inserted column

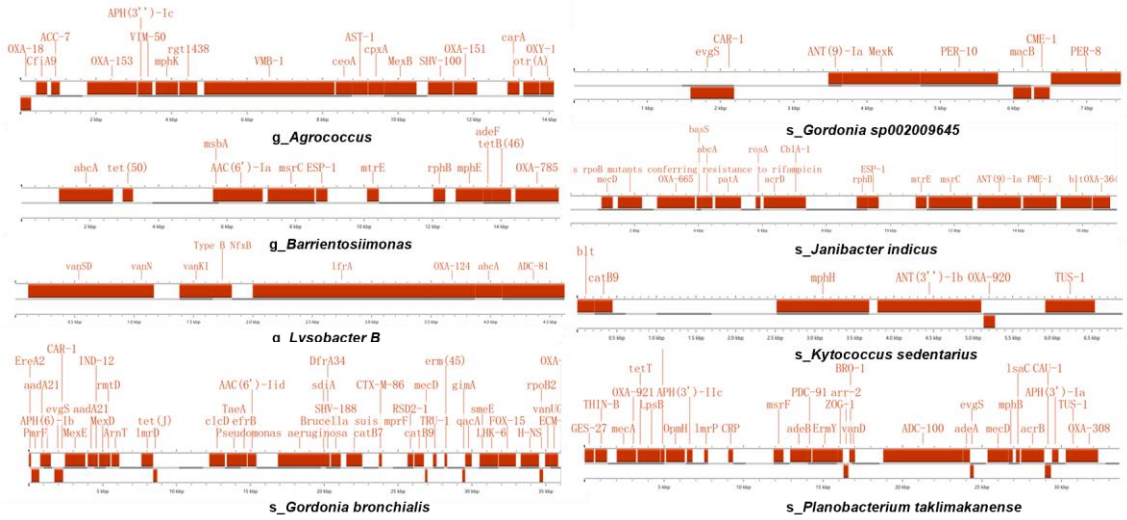

Figure 9. ARGs in insertion sequence identified in rMAGs. *Red boxes represent gene segments of ARGs of different lengths on contigs with predicted gene names.*

Discussion

In residential environments, the hand is the conduit for the frequent exchange of microorganisms between the environment and the body. In this study, longitudinal experiments allowed us to analyze microbiota species composition and diversity transmission patterns over time provide insights into the carriage of MGEs and ARGs during microbial transmission.

First, *Gordonia bronchialis*, *Janibacter indicus* and *Micrococcus luteus* were determined to be the three most prevalent bacteria on palms and doorknob. All of them are bacteria typically present in the environment and on the skin (Ramanan et al., 2013, Young et al., 2010, Tannock, 1995). Interestingly, on the sixth day, the relative abundance of certain species on the hands and doorknobs of the subjects changed significantly, particularly *Gordonia bronchialis* and *Janibacter indicus*. It unclear currently needs to be determined under what conditions the two species compete or if they occupy the same ecological niche. Additional research is required to determine the competitive relationship between these two bacteria for resources. Pathogens with strong skin-surface dispersal ability, such as common skin bacteria, were also screened.

Identification at the genome level revealed the presence of nine microorganisms on both palms and doorknob. In conjunction with the microbial homogeneity suggested by the previous similarity results, it is possible to hypothesize that these bacteria were transmitted between the hand and the doorknob. On the basis of the order of appearance of their MAGs, it is possible that these bacteria were initially introduced into residential environments via palms and doorknob. However, the mutual transmission strength between the two types of surfaces could not be determined. In previous studies, researchers calculated the frequency

of bacterial transfer from occupants to domestic surfaces or household surfaces to occupants using dynamic Bayesian network models (Lax *et al.*, 2014). However, a surrogate source of microorganism was required by this model to predict the microbial transmission potentials from outdoor to indoor environments, limiting the assessment of the transmission strength and direction via skin-surface touch within the indoor spaces or by other routes with microorganisms from unknown sources.

In recent years, the overuse of antibiotics has accelerated the evolution of antibiotic-resistant bacteria and the dissimilation of ARGs in the environment. Genome-level analysis revealed that bacteria that simultaneously appeared on the hand contained resistance proteins for the ABC antibiotic efflux pump, the MFS antibiotic efflux pump, and the RND antibiotic efflux pump. They are all members of the primary superfamily of bacterial drug efflux transporters (Méndez and Salas, 2001, Saidijam et al., 2006, Daniels and Ramos, 2009). Antibiotic efflux pumps are believed to substantially contribute to acquired bacterial resistance, as they facilitate the efflux of antibiotics from the intracellular to the extracellular environment (Van Bambeke et al., 2000, Webber and Piddock, 2003). Pathogenic bacteria are thought to acquire an advantage over antimicrobial agents by transferring resistance gene determinants between different bacterial species and acquiring a resistant phenotype to the majority of antimicrobial agents (Serwecińska, 2020). ARGs were identified from IS on the skin surface and found to have spread throughout the genome. It has been observed that bacteria perform ARG transfer with a significantly higher frequency sing MGEs such as plasmids, transposons, and phages (Paul and Das, 2022). Therefore, these species harboring both MGEs or IS and ARGs might have a high HGT potential and may pose a risk for the transmission of genomes associated with

environmental resistance to humans. Nevertheless, due to the short length of IS, ARGs on IS may be less accurate, necessitating more precise analytical tools.

Multiple factors influence the transmission of microorganisms between occupants and the built environment. The hand microbiome is dynamic and is affected by personal activities such as hand cleansing frequency and oral antibiotic use (Edmonds-Wilson et al., 2015, Fierer et al., 2008). However, in this study, subject characteristics such as occupation, hand-washing frequency, and hygiene habits were not measured or considered. In addition, the smaller sample size reduces the reliability of conclusions drawn from multisite studies conducted over extended periods. Nevertheless, understanding the microbial transmission potentials via skin-surface touch could provide a foundation for developing intervention strategies for reducing the adverse effects of disease transmission and enhancing the health and well-being of the occupants and cohabitants. In addition, the investigation and analysis methods used in this study can be applied to research in hospital settings to reveal the possible microbial colonization patterns, the occurrence pattern and transmission routes of opportunistic pathogens or multi-drug resistant bacteria, and to provide hospitals with recommendations to prevent infection and protect patients.

Conclusion

In conclusion, the exchange of microbiota between humans and indoor environments is a complex phenomenon that may have health implications for humans. Through longitudinal metagenomic analysis, this study provides insights into the transmission of microbial species and ARGs between humans and residential surfaces, highlighting the significance of investigating the diffusion of ARGs in indoor environments during

microbiota exchange. More research is required to thoroughly define more precise bidirectional microbial transmission patterns and their implications for human health, and develop effective strategies to control the spread of antibiotic resistance in indoor environments, particularly in high-risk settings such as hospitals and nursing homes.

List of Abbreviations

Abbreviation	Definition
ARG	Antibiotic Resistance Gene
MGE	Mobile Genetic Element
HGT	Horizontal Gene Transfer
IS	Insertion Sequences
MAG	Metagenomic Assembled Genome
rMAG	Representative Metagenomic Assembled Genome
PCoA	Principal Coordinates Analysis
BCD	Bray-Curtis Dissimilarity
ABC	ATP_Binding Cassette
MFS	Major Facilitator Superfamily
RND	Resistance-Nodulation-Cell Division

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Appendix

Appendix Table 1. Pathogens predicted to be highly dispersive by the Sloan neutral community model

Species	P value	Frequency	Predicted frequency	Partition	Mean abundance
Burkholderia lata	3.32E-06	1	0.9436	Above	3.32E-06
Gardnerella vaginalis	4.44E-06	1	0.9889	Above	4.44E-06
Aeromonas veronii	4.69E-06	1	0.9926	Above	4.69E-06
Bordetella holmesii	4.89E-06	1	0.9947	Above	4.89E-06
Acinetobacter lwoffii	5.25E-06	1	0.9971	Above	5.25E-06
Mycobacterium tuberculosis	6.24E-06	1	0.9995	Above	6.24E-06
Staphylococcus pettenkoferi	6.46E-06	1	0.9997	Above	6.46E-06
Acinetobacter schindleri	7.02E-06	1	0.9999	Above	7.02E-06
Clostridium botulinum	7.17E-06	1	0.9999	Above	7.17E-06
Mycolicibacterium fortuitum	7.20E-06	1	0.9999	Above	7.20E-06
Legionella pneumophila	7.33E-06	1	0.9999	Above	7.33E-06
Aeromonas hydrophila	7.66E-06	1	1.0000	Above	7.66E-06
Burkholderia vietnamiensis	9.03E-06	1	1.0000	Above	9.03E-06
Acinetobacter johnsonii	9.05E-06	1	1.0000	Above	9.05E-06
Acinetobacter junii	9.41E-06	1	1.0000	Above	9.41E-06
Neisseria cinerea	1.04E-05	1	1.0000	Above	1.04E-05
Pasteurella multocida	1.09E-05	1	1.0000	Above	1.09E-05
Staphylococcus cohnii	1.10E-05	1	1.0000	Above	1.10E-05
Bordetella trematum	1.13E-05	1	1.0000	Above	1.13E-05
Fusobacterium varium	1.42E-06	0.9375	0.5387	Above	1.42E-06
Bacillus subtilis	1.67E-06	0.9375	0.6298	Above	1.67E-06
Bacillus pumilus	2.21E-06	0.9375	0.7816	Above	2.21E-06
Acinetobacter bereziniae	1.12E-06	0.875	0.4169	Above	1.12E-06
Morganella morganii	1.27E-06	0.875	0.4792	Above	1.27E-06
Clostridioides difficile	1.37E-06	0.875	0.5193	Above	1.37E-06
Aeromonas caviae	1.22E-06	0.8125	0.4587	Above	1.22E-06
Serratia plymuthica	1.45E-06	0.8125	0.5483	Above	1.45E-06

Francisella tularensis	1.34E-06	0.75	0.5094	Above	1.34E-06
Chlamydia trachomatis	7.86E-07	0.6875	0.2773	Above	7.86E-07
Helicobacter pylori	9.89E-07	0.6875	0.3635	Above	9.89E-07
Campylobacter sputorum	5.07E-07	0.625	0.1630	Above	5.07E-07
Bacteroides caccae	8.62E-07	0.625	0.3096	Above	8.62E-07
Enterobacter bugandensis	8.87E-07	0.625	0.3204	Above	8.87E-07
Bordetella pertussis	9.38E-07	0.625	0.3420	Above	9.38E-07
Providencia rettgeri	9.89E-07	0.625	0.3635	Above	9.89E-07
Enterococcus faecium	1.01E-06	0.625	0.3743	Above	1.01E-06
Acinetobacter oleivorans	2.79E-07	0.5625	0.0797	Above	2.79E-07
Enterobacter roggenkampii	5.32E-07	0.5625	0.1729	Above	5.32E-07
Vibrio furnissii	4.56E-07	0.5	0.1434	Above	4.56E-07
Bordetella parapertussis	5.07E-07	0.5	0.1630	Above	5.07E-07
Bacteroides zoogleoformans	5.32E-07	0.5	0.1729	Above	5.32E-07
Prevotella ruminicola	7.10E-07	0.5	0.2453	Above	7.10E-07
Brucella melitensis	3.04E-07	0.4375	0.0883	Above	3.04E-07
Brucella abortus	4.06E-07	0.4375	0.1244	Above	4.06E-07
Shewanella putrefaciens	4.82E-07	0.4375	0.1531	Above	4.82E-07
Clostridium tetani	2.28E-07	0.375	0.0632	Above	2.28E-07
Listeria innocua	2.03E-07	0.3125	0.0552	Above	2.03E-07